MILITARY SPECIFICATION

PACKING, PREFORMED, STRAIGHT THREAD TUBE FITTING BOSS, TYPE II HYDRAULIC (-65°F to +275°F)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of elastomeric straight thread tube fitting boss packings for use in 3000 psi hydraulic systems using MIL-H-5606 fluid throughout a temperature range of -65°F to +275°F.

2. APPLICABLE DOCUMENTS

2.1 Government documents normally furnished - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250/12	Aluminum Alloy 7075, Plate and Sheet
QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet
QQ-B-613	Brass, Leaded and Non-Leaded, Plate, Rolled Bar, Sheet and Strip
QQ-B-750	Bronze, Phosphor, Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-4861	Packing, Preformed, Rubber, Packing, Packaging of
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-H-6083	Hydraulic Fluid, Petroleum Base, Preservative
MIL-S-18729	Steel Plate, Sheet and Strip, Alloy, 4130, Aircraft Quality

FSC 5330

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STANDARDS

Federal

FED-STD-601	Rubber, Sampling and Testing
FED-STD-791	Lubricants, Liquid Fuels, and Related Products, Methods of Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-413	Visual Inspection Guide for Rubber O-Rings
MS 3393	Packing, Preformed, Straight Thread Tube Fitting Boss, Hydraulic (-65°F to +275°F)
MS 33649	Bosses, Fluid Connection - Internal Straight Thread

Air Force - Navy Aeronautical

AN814	Plug and Bleeder - Screw Thread
AN937	Cross-Internal Screw Thread

Air Force - Navy Aeronautical Design

AND10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread
AND10064	Fittings Installation of Flared Tube Straight Threaded Connectors

PUBLICATIONS

Air Force - Navy Aeronautical Bulletin

No. 438	Age Controls for Synthetic Rubber Parts

(When requesting applicable documents, refer to both title and number. Copies of unclassified documents may be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Requests for copies of classified documents should be addressed to the Naval Publications and Forms Center, via the cognizant Government inspector.)

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2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials Standard

ASTM-D91	Method of Test for Precipitation Number of Lubricating Oils
ASTM-D1414	Testing Rubber O-Rings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Society of Automotive Engineers Aerospace Material Specification

AMS 4029	Sheet and Plate 2014T6

(Applications for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Ave., New York, New York 10017.)

USA Standards Institute

USAS B46.1 - 1962	Surface Texture

(Applications for copies should be addressed to The American Standards Association, 10 East 40th St., New York, New York 10016.)

3. REQUIREMENTS

3.1 First article - The packing furnished under this specification shall be a product which has been inspected and has passed the first article inspection specified herein. (See 6.2.)

3.2 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials - The MS3393 packings shall be fabricated from an elastomeric material suitable for use in hydraulic fluid conforming to MIL-H-5606. The material shall contain no substances which, when in contact with the fluid will adversely affect the properties of the fluid or packing.

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3.4 Design and construction - The design and construction of the packing shall be in accordance with MS3393. The packings shall be made from continuous-circle preforms only.

3.5 Performance - The packing and batch material shall meet the performance requirements specified in Section 4 when subjected to the applicable tests.

3.6 Age - Unless otherwise specified, the age limitations for the finished product shall conform to ANA Bulletin No. 438.

3.7 Finish -

3.7.1 Trimming - Mold flash shall be removed from the packings in such a manner that they conform to the requirements specified herein and in MS3393. The method used to trim the mold flash shall not remove the surface of the packing to a greater extent than would result from a light, nonabrasive buffing.

3.7.2 Surface - All surfaces of packings shall be free from defects, blemishes, or irregularities other than minor pits, raised spots, or tool marks resulting exclusively from mold imperfections (not from dirt in the mold) and which are not greater than 0.003 inch in height or depth from the packing surface. Such blemishes shall not be grouped closely and shall cover not more than 10 percent of the surface.

3.8 Workmanship - The packing shall be free of unbonded or weak sections caused by internal mold seams, cold shuts, foreign inclusions or other detrimental characteristics. The packing shall be uniform in quality and condition; clean, smooth, and free from foreign materials or defects detrimental to appearance or performance. Major defects as defined in MIL-STD-413 shall be cause of rejection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection - The inspection of packings shall be classified as follows:

(a) First article inspection (see 4.3)

(b) Quality conformance inspection (see 4.4)

4.3 First article inspection - First article inspection shall consist of all of the inspections of this specification conducted in the order listed under 4.6 and in Table I.

4.3.1 Sampling for first article inspection - First article test samples shall be as specified in Table I and shall include the following:

(a) 24 MS3393 - 3 packings

(b) 4 MS3393 - 4 packings

(c) 4 MS3393 - 8 packings

(d) 4 MS3393 - 12 packings

(e) 58 MS3393 - 16 packings

(f) 3 ASTM hardness discs (1/4-inch thick, with a 1.0-in. diameter)

4.3.1.1 Hardness discs shall be formulated, processed, and cured the same as the accompanying packing. The packings shall represent the manufacturer's production product, particularly with respect to stock preparation, mold design, mold finish, and molding technique.

4.4 Quality conformance inspection - Quality conformance inspection shall consist of sampling inspections only. Conducted in the order listed in Table II.

4.4.1 Sampling inspection - A random sample shall be selected from each inspection lot or batch in accordance with Table II and shall be subjected to the inspections therein.

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TABLE I

First Article Inspection Tests and Samples 1/							
Physical Property Tests	Paragraph Number	Specimen					
Unaged							
Hardness	4.6.2.1	1 ASTM Hardness disc (1/4-inch thick x 1.0-in. diameter)					
Tensile strength	4.6.2.2	2 MS3393-16					
Elongation	4.6.2.3	2 MS3393-16					
Specific gravity	4.6.2.4	3 MS3393-16					
Permanent set	4.6.2.5	2 MS3393-3					
Temperature retraction	4.6.2.6	3 MS3393-16					
Corrosion & adhesion	4.6.2.8	12 MS3393-16					
Air Aged							
Hardness	4.6.2.1	1 ASTM Hardness disc (1/4-inch thick x 1.0-in. diameter)					
Tensile strength	4.6.2.2	2 MS3393-16					
Elongation	4.6.2.3	2 MS3393-16					
Permanent set	4.6.2.5	2 MS3393-3					
Oil Aged at 158°F							
Temperature retraction	4.6.2.6	3 MS3393-16					
Oil Aged at 275°F							
Hardness	4.6.2.1	1 ASTM Hardness disc (1/4-inch thick x 1.0-in. diameter)					
Tensile strength	4.6.2.2	2 MS3393-16					
Elongation	4.6.2.3	2 MS3393-16					
Permanent set	4.6.2.5	2 MS3393-3					
Volume change	4.6.2.7	3 MS3393-16					
Performance Tests		MS3393 Packing					
		-3 Size	-4 Size	-8 Size	-12 Size	-16 Size	
Installation stretch at normal temperature	4.6.2.9.1	12					
Installation stretch at -20°F	4.6.2.9.2	6					
Crush	4.6.2.10					9	
Extrusion	4.6.2.11					9	
Impulse	4.6.2.12		4	4	4	4	

1/ All MS3393 packings shall be examined in accordance with 4.6.1 before the initiation of first article tests.

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TABLE II

Quality Conformance Inspection (all specimens unaged except Volume change)				
Inspection	Paragraph Number	Sampling plan per	Inspection level	AQL (percent defective)
Examination of product Examination of preparation for delivery 1/	4.6.1 4.6.1.1	MIL-STD-105 MIL-STD-105	S-4 S-4	4.0 6.5
		Specimen		
Batch 2/, 3/				
(a) Hardness	4.6.3.1	1 ASTM compression molded hardness disc (1/4-in. thick x 1.0-in. diameter)		
(b) Tensile strength	4.6.3.2	2 MS3393-16 packings		
(c) Elongation	4.6.3.3	2 MS3393-16 packings		
(d) Specific gravity	4.6.3.4	3 MS3393-16 packings		
Finished product 3/, 4/				
(a) Tensile strength	4.6.3.2	2 MS3393-16 packings		
(b) Elongation	4.6.3.3	2 MS3393-16 packings		
(c) Specific gravity	4.6.3.4	3 MS3393-16 packings		
Process monitoring 3/, 5/				
(a) Permanent set	4.6.3.5	2 MS3393-3 packings		
(b) Temperature retraction	4.6.3.6	3 MS3393-16 packings		
(c) Volume change	4.6.3.7	3 MS3393-16 packings		
(d) Corrosion and adhesion	4.6.3.8	12 MS3393-16 packings		
(e) Crush	4.6.2.10	3 MS3393-16 packings		

1/ Except as required by specifications referenced in Section 5.

2/ A batch shall consist of the quantity of material compounded at any one time and ready for molding and curing into a finished product. Test specimen shall be molded from each batch and finished with the same state of cure as the finished product.

3/ If a test specimen or sample fails to pass any of these tests, the lot or batch represented by the unit shall be rejected.

4/ Finished product tests shall be performed on unaged specimens or samples from each day's production molded concurrently with other production parts.

5/ Process monitoring test shall be performed at 3 month intervals on unaged specimen molded concurrently with production parts, and using the same production techniques.

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4.5 Inspection conditions -

4.5.1 Ambient (room or normal) temperature and pressure - Unless otherwise specified, tests shall be conducted at an ambient temperature of 75 ±5°F and at an ambient barometric pressure of 28 to 32 inches of mercury. If tests are conducted at different ambient conditions, appropriate correction factors shall be applied to the pressure readings.

4.5.2 Air aging - When required for physical properties tests, air aging shall be in accordance with FED-STD-601, Method 7221, except that specimens shall be aged for 166 ± 1 hours at 158 ± 2°F.

4.5.3 Oil aging - When required for physical properties tests, oil aging of test specimens shall be as follows. The fluid and samples to be aged shall be placed in a non-pressurized closed container or reflux condenser. The test fluid shall not be in contact with any part of the container cover. The packings shall be suspended in the fluid by a glass or nichrome wire rack and shall not come in contact with any part of the container. A new batch of fluid shall be used for each aging test. In all cases the ratio of quantity of fluid to the weight of the seals or material being aged shall be 20 ml of fluid for every gram of material. Seals or materials shall be loosely separated in the fluid.

4.5.3.1 Oil aging at 158°F for temperature retraction (4.6.2.6) and extrusion (4.6.2.11) - Aging (fluid) temperature shall be maintained at 158 ± 2°F for a period of 166 ± 1 hours.

4.5.3.2 Oil aging at 275°F for all other oil aging tests - Aging (fluid) temperature shall be maintained at 275 ± 10°F for a period of 72 hours.

4.5.4 Test fluid - Unless otherwise specified the fluid used for all applicable tests under 4.6 shall conform to MIL-H-5606.

4.6 Inspection methods -

4.6.1 Examination - The sample packings shall be thoroughly examined to determine conformance with this specification and applicable drawings for finish, trimming, marking, cross-sectional dimensions, inside and outside diameters, and all other requirements not covered by tests.

4.6.1.1 Examination of preparation for delivery - The packings shall be examined to ascertain that the preparation for delivery conforms to requirements of Section 5 of this specification.

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4.6.2 First article -

4.6.2.1 Hardness - This test shall be in accordance with FED-STD-601, method 3021, using ASTM hardness disc specimens. The test shall be conducted at 75 ± 5°F, and the Shore A durometer hardness value for the unaged specimen shall be not less than 75 nor more than 85 points. The hardness value for the air aged specimen (see 4.5.2) shall be within -0 and +5 points of the unaged specimen. The hardness value for the oil aged specimen (see 4.5.3 and 4.5.3.2) shall be within ± 5 points of the unaged specimen.

4.6.2.2 Tensile strength - This test shall be in accordance with ASTM D 1414 using two MS3393-16 packings each for the unaged, air aged, and oil aged tests. The tensile strength for the unaged specimen shall not be less than 1500 psi. The tensile strength values for the air aged specimen (see 4.5.2) and oil aged specimen (see 4.5.3 and 4.5.3.2) shall be at least 80 percent of the value obtained for the unaged specimen.

4.6.2.3 Elongation - This test shall be in accordance with ASTM D 1414 using two MS3393-16 packings each for the unaged, air aged, and oil aged tests. The elongation for the unaged specimen shall not be less than 125 percent. The elongations for the air aged specimen (see 4.5.2) and oil aged specimen (see 4.5.3 and 4.5.3.2) shall be at least 70 percent of the value obtained for the unaged specimen.

4.6.2.4 Specific gravity - This test shall be in accordance with FED-STD-601, method 14011, except that three MS3393-16 packings shall be used and the following paragraphs of method 14011 shall not apply.

Section 2 - Specimen - rubber content

Section 4 - Specific gravity for rubber content

Section 5 - Results - rubber content

The specific gravity shall be not less than 1.25 nor more than 1.45.

4.6.2.5 Permanent set - Permanent set tests shall be conducted using a suitable tapered cone and two MS3393-3 packings each for the unaged, air aged, and oil aged tests. The original inside diameter of the packing shall be measured and the packing then shall be stretched over a tapered cone lubricated with MIL-H-5606 fluid until the inside diameter of the sealing surface has been increased 35 percent. The packing shall be held in the stretched state for 10 minutes, removed from the cone, and the new inside diameter determined after waiting 10 minutes. Permanent set (P.S.) shall be calculated as follows:

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$$P.S.(\%) = \frac{\text{(I.D. stretched packing in inches)-(I.D. unstretched packing in inches)}}{\text{(I.D. unstretched packing in inches)}} \times 100$$

There shall be no evidence of breaking or cracking of the test samples. The permanent set for unaged, air aged (see 4.5.2), and oil aged (see 4.5.3 and 4.5.3.2) samples shall not exceed 3.0 percent.

4.6.2.6 Temperature retraction - This test shall be in accordance with ASTM D 1414, except that three MS3393-16 packings each for the unaged and oil aged tests shall be used and shall be tested in a test set-up similar to Figure 1. The rate of warming shall be 1°F per minute. The temperature retraction for both unaged and oil aged (see 4.5.3 and 4.5.3.1) samples at 50 percent elongation and 10 percent return (50/10 reading) shall occur at -40°F or lower.

4.6.2.7 Volume change - The volume change of the test sample shall be determined using three MS3393-16 packings and shall be conducted in accordance with FED-STD-601, method 6211, after oil aging in accordance with 4.5.3 and 4.5.3.2. The volume of the test sample shall not decrease and shall not increase in excess of 12.0 percent.

4.6.2.8 Corrosion and adhesion - Twelve MS3393-16 packings shall be prepared for corrosion testing by prehumidifying the packings in a desiccator or similar humidity chamber, maintained at 92 percent minimum relative humidity and room temperature for 72 hours. Metal plates used shall be polished to a surface roughness of 4 to 16 microinches RHR finish in accordance with USAS B46.1-1962. The edges shall also be polished to reduce the formation of edge corrosion. Metal plates shall be washed with precipitation naphtha, as specified in Method 3101 of FED-STD-791, ASTM D91-61, or similar degreasing agent. The metal plates used shall be as follows (a set of seven control plates, one of each of the metals tested shall be included):

Aluminum alloy	QQ-A-250/4, Condition T6
Aluminum alloy	QQ-A-250/12, Condition T651 Heat Treated to T73
Aluminum alloy	AMS 4029, 2014 Condition T6
Brass	QQ-B-613, Composition 4
Phosphorbronze	QQ-B-750, Composition A
Steel	MIL-S-18729, Condition N
Steel	Cadmium plated per QQ-P-416

The humidified packings and the metal plates shall be immersed in MIL-H-6083, type 1 fluid and drained to the drip point. The packings and plates shall then be laid together in a stack such that at least two packings contact each specified metal. The stack shall be held together with a force of 20 to 30 pounds and placed in a desiccator which is maintained at 90 to 92 percent relative humidity

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at room temperature. A separate set of metallic plates shall also be prepared (buffed, cleaned and dipped in MIL-H-6083, type 1 fluid, and drained) and placed in the desiccator in such a manner that the control plates shall not touch each other or any of the packings (dipotassium acid phosphate, K_2HPO_4, when placed in distilled water in sufficient quantity to produce a saturated solution, will maintain approximately 92 percent relative humidity in a sealed desiccator at 68°F).

Time of storage for this portion of test shall be 336 hours. No more than 15 minutes shall elapse between the time the test samples are removed from the prehumidifying chamber and placed in the stacked condition in the second humidity chamber.

At the termination of this test, the test samples shall be examined as follows:

(a) The surfaces of the plates that were in contact with the packings shall be inspected for evidence of deposit formation or adhesion of packing material. If deposits or adhesion are evidenced the surfaces of the plates shall be washed in precipitation naphtha. Deposits determined as rubber compounds or elements therefrom, which can be removed by this process and which do not occur on the separate control plates, shall be construed as adhesion. There shall be no evidence of adhesion of the packing material to the metal plates.

(b) If any other marks remain on the surfaces of the plates after step (a) the surfaces shall be lightly polished with a non-abrasive buffing cloth. Any pits or eroded marks remaining after this process shall be construed to be corrosion. Discoloration or staining (marks which do not physically affect the surface of the plates and which easily wash or buff off) shall not be considered detrimental. If any doubt should arise about the presence of pitting, erosion or corrosion on the metal plates, a microscope of approximately 10 to 15-power magnification shall be used to determine the actual condition. There shall be no evidence of severe discoloration, corrosion (pitting), or erosion.

4.6.2.9 Installation stretch -

4.6.2.9.1 Installation stretch at normal temperature - Twelve MS3393-3 packings shall be used in this test. Six packings shall be unaged and six shall be air aged (see 4.5.2) unless the result of the elongation test indicated that the material decreased in elongation due to air aging by more than 30 percent, in

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which case all twelve packings shall be air aged. The packings shall be stretched onto a smooth metallic mandrel of 0.390 ±.002 inch diameter and allowed to remain in this condition on the 0.390 inch diameter section of the mandrel for 1 hour at room temperature. There shall be no evidence of breakage or cracking of any of the packings.

4.6.2.9.2 Installation stretch at -20°F - The test of 4.6.2.9.1 shall be duplicated at -20 ± 2°F except that three each of the unaged and air aged (see 4.5.2) packings shall be used (regardless of change in elongation). The unaged and air-aged packings and the threaded mandrel (see Figure 2) shall be held at -20 ± 2°F for 24 hours (packings unstretched) prior to stretching on the mandrel. Each packing shall be rapidly pushed from the cone of the mandrel over the threaded 0.390 inch diameter onto the smooth 0.320 inch diameter undercut. There shall be no breakage of packings in this test and each packing shall decrease in inside diameter to grip the 0.320 inch diameter undercut in a maximum of 10 seconds after being inserted on the mandrel while still at the specified temperature.

4.6.2.10 Crush - Nine MS3393-16 packings shall be used in this test. Three shall be unaged, three shall be air aged (see 4.5.2), and three shall be oil aged (see 4.5.3 and 4.5.3.2). The packing shall be tested in a test setup similar to Figure 3. The packings shall each be twisted (inside out) one-half turn about the cross section and laid between rigid steel plates which have a surface finish of 40 to 60 microinches obtained by sandblasting a commercial grind finish as suggested in Figure 3. The air-aged packings shall be laid together in one layer, the oil aged in a second layer and the unaged in a third layer. The stack shall then be subjected to a constant axial compression load of 16,800 pounds, or 5600 pounds for each packing in a layer (number of layers does not affect the applied load). The stack shall be left in the compressed state for 68 hours and then each ring shall be removed and inspected immediately for cracking or disintegration. The packings shall be stored at room temperature in a free condition and the permanent set of seal cross-section determined at the end of 20 hours. No cracking or disintegration shall be evident. The cross sectional area of the packing shall not have been reduced in permanent set more than 25 percent of the "as manufactured" size.

4.6.2.11 Extrusion - Nine MS3393-16 packings shall be used for this test. Three shall be used unaged, three air aged (see 4.5.2), and three oil aged (see 4.5.3 and 4.5.3.1). The nine packings shall be installed one each on AN814-16 fittings, and the fittings shall then be installed in AND10050 bosses of applicable size (AN937-16 boss fittings may be used). The fittings shall be screwed into the bosses so that an 0.012 inch feeler gage can be inserted between the flat boss-face and the adjacent face of the AN814 fitting (this allows 0.012 inch extrusion gap around the test packings). Hydrostatic pressure of 3000 ± 100 psi shall be maintained constantly on the inside of the AN814 fittings and bosses, and the assemblies shall be maintained at 275 ± 2°F for 72 hours. During this period there shall be no evidence of excessive extrusion of packing or leakage of fluid, and at the termination of the test the packings shall show no cracking, disintegration or other failure. The packings may remold to the shape of the confining cavity. A small

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undeteriorated ridge at the extrusion point of the packing shall not be considered cause for rejection.

4.6.2.12 Impulse - Standard MS3393 packings in quantities and sizes specified in Table I shall be used for this test. The unaged packings shall be installed on standard fittings in accordance with AND10064, sheets 2 and 3, which shall, in turn, be installed in a test fixture as shown in Figure 4. The test fixture shall be installed in a test set-up similar to Figure 5. Impulse pressure (see Figure 6) shall be applied at a rate of 35 ± 10 cycles per minute through the range of 0 to 3000 psi with a peak pressure of 150 percent of 3000 psi at the start of each pressure cycle for a total of one million cycles. There shall be no leakage at any time when the packing assembly is subjected to the following schedule (shown graphically on Figure 7):

(a) Conduct a static leak check at 5 psi, 3000 psi, and 4500 psi.
(b) 50,000 cycles at room temperature, followed by a static pressure leak check at 5 psi, 3000 psi, and 4500 psi.
(c) 50,000 cycles at -65 ± 5°F, followed by static leak check.
(d) Static leak check at room temperature.
(e) Static leak check at 275 ± 5°F.
(f) 200,000 cycles at 275°F, followed by static leak check at 275°F, room temperature and -65°F.
(g) 200,000 cycles starting at -65°F and increasing temperature to +275°F followed by static leak check at 275°F, room temperature and -65°F.
(h) Repeat paragraph (g) until a total of 1,100,000 impulse cycles have been completed.

4.6.3 Quality conformance - All specimens used in quality conformance inspection shall be unaged (except those used for Volume change).

4.6.3.1 Hardness (batch) - Test to be same as 4.6.2.1 except that the hardness value shall not vary more than ± 5 Shore A durometer points from the value obtained for the unaged specimen in 4.6.2.1.

4.6.3.2 Tensile strength (batch and finished product) - Test to be same as 4.6.2.2 except that the tensile strength shall not vary more than ± 15 percent from the value obtained for the unaged specimens in 4.6.2.2.

4.6.3.3 Elongation (batch and finished product) - Test to be same as 4.6.2.3 except that the elongation shall not vary more than ±20 percent from the value obtained for the unaged specimen in 4.6.2.3.

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4.6.3.4 Specific gravity (batch and finished product) - Test to be same as 4.6.2.4 except that the specific gravity shall not vary more than ±0.02 from the value obtained in 4.6.2.4.

4.6.3.5 Permanent set (process monitoring) - Test to be same as 4.6.2.5 except that the permanent set shall not exceed 5.0 percent.

4.6.3.6 Temperature retraction (process monitoring) - Test to be same as 4.6.2.6 except that the temperature retraction shall not vary more than ±2°F from the value obtained for the unaged sample in 4.6.2.6.

4.6.3.7 Volume change (process monitoring) - Test to be same as 4.6.2.7 except that the volume change of the oil aged packing shall be within ±3 percent of the value obtained in 4.6.2.7, and the test sample shall not shrink.

4.6.3.8 Corrosion and adhesion (process monitoring) - Test to be same as 4.6.2.8 except that the storage time for the test samples in the desiccator for the final portion of the test shall be 96 hours instead of 336 hours.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking - Unit packaging, intermediate packaging, and packing shall be in accordance with Level A of MIL-P-4861. Preparation and marking shall be in accordance with MIL-P-4861 (see 6.2.b).

6. NOTES

6.1 Intended use - The packings are intended for use in type II class 3000-psi aircraft hydraulic systems to seal standard tube fittings in bosses as indicated in AND 10064 (sheets 2 and 3).

6.2 Information for contracting officer - Contracts or orders should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing required (see 5.1).
(c) Packings subjected to tests are not to be considered or shipped as part of the contract or order.
(d) Items of data required (see 6.3).

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6.2.1 Contracts or orders should specify the following provisions for first article inspection.

6.2.1.1 Whether first article inspection is required – When a contractor is in continuous production of the packings from contract to contract, consideration should be given to waive the first article inspections. If inspection is required, indicate.

(a) where the first article inspection is to be conducted (at the contractor's plant or Government or commercial laboratory).

(b) that the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification, or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order.

6.3.1 First article data - When first article samples are submitted (see 6.2), they should be accompanied by a complete inspection report showing the results of the contractor's inspections. The inspection report should include the following·

(a) Report of all inspections graphically presented when possible, together with a detailed statement indicating compliance or extent of noncompliance with all requirements of this specification and applicable military standard referring specifically to paragraph numbers. Wherever a requirement is considered to be not applicable, the report should so state.

(b) Diagrams of inspection set-ups. A complete description of inspection equipment and inspection procedures.

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(c) Reproducible outline and description of inspection conditions. Where inspections specified in this specification are not considered applicable, the reason, and the substituted inspection should be clearly described.

(d) Copies of inspection log sheets.

(e) Photographs when available.

6.4 International standardization agreements - Certain provisions (see 3.4) of this specification are the subject of international standardization agreement (ASCC AIR STD 17/21 Pipe Fitting Sealing Rings). When amendment, revision, or cancellation of this specification is proposed, which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices if required.

Custodians:

Army - MR
Navy - AS
Air Force - 11

Preparing Activity:

Navy - AS

Review Activities:

Army - MR, MO, WC
Navy - WP
Air Force - 11, 82

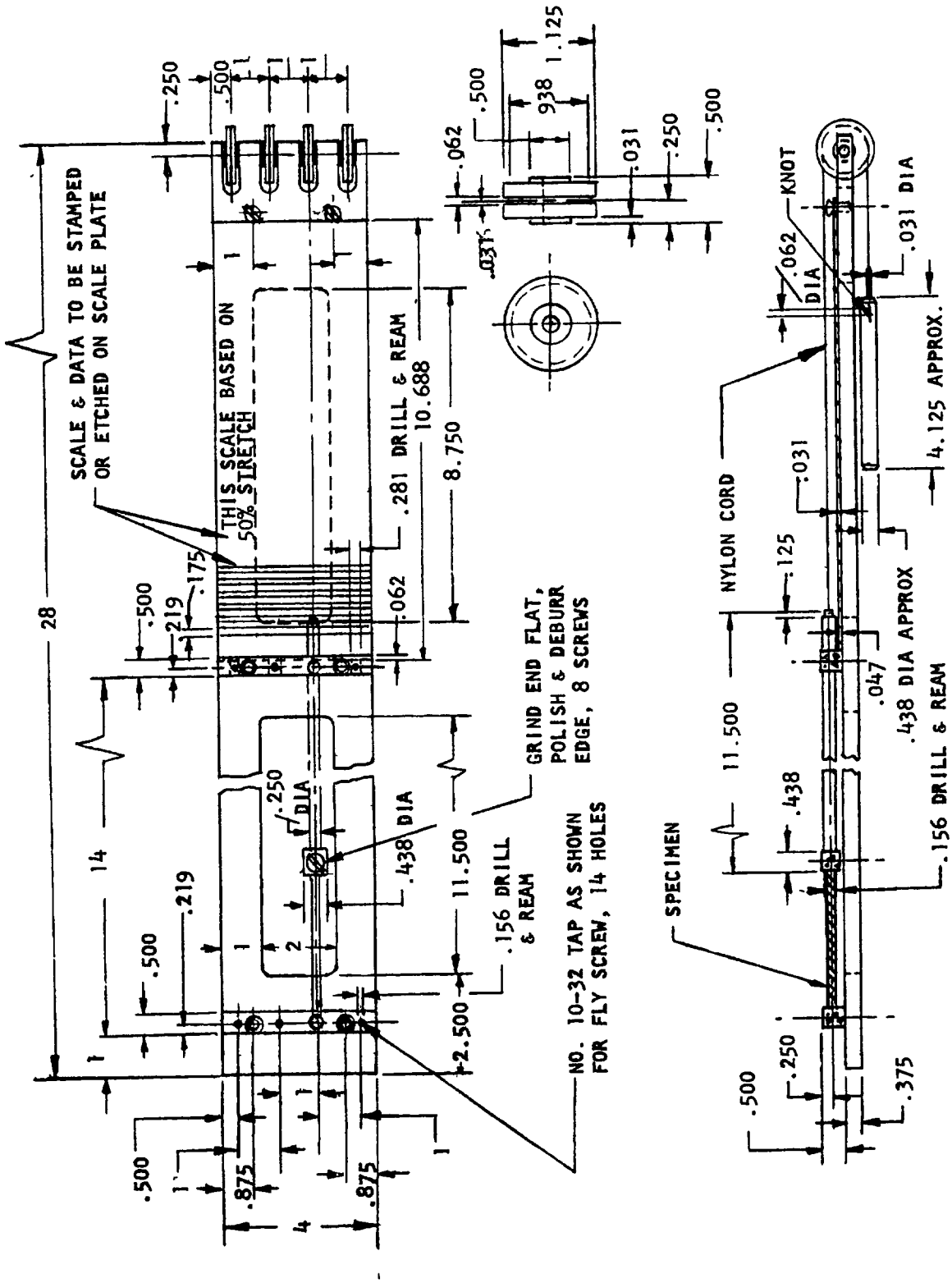
User Interest:

Army - MI
Navy - SH
Air Force -

(DoD Project No. 5330-0309)

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DoD Standardization Documents.

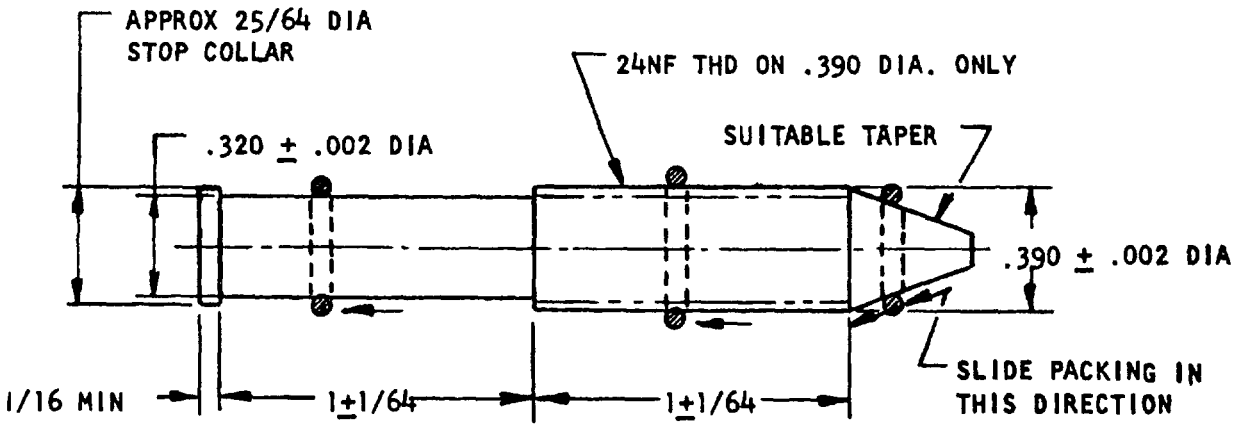
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FINISH: PLATE WITH COPPER THEN HARD CHROME ALL OVER.
DIMENSIONS IN INCHES.

FIGURE 1. - TEMPERATURE RETRACTION APPARATUS.

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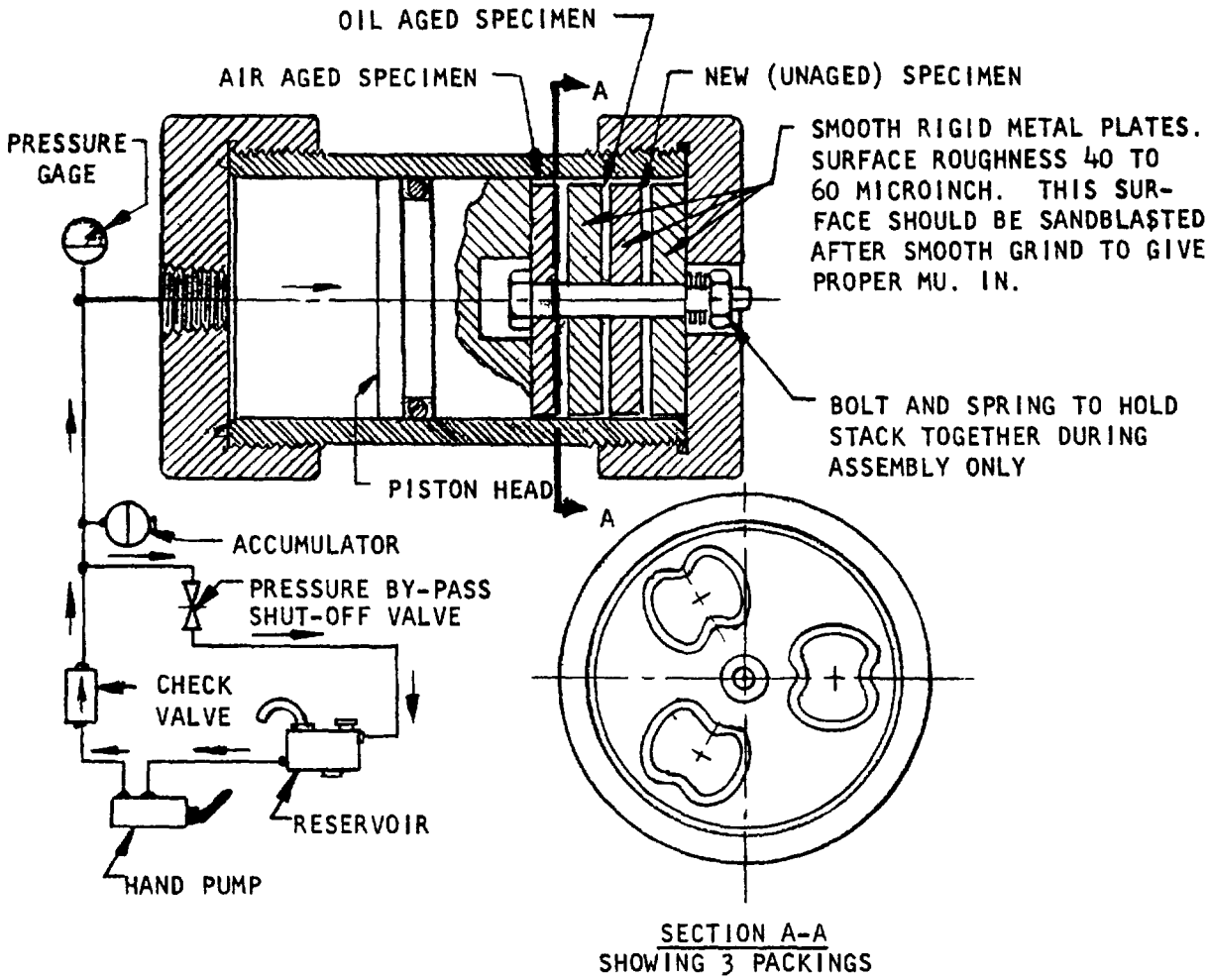


DIMENSIONS IN INCHES

FIGURE 2

Installation stretch test mandrel

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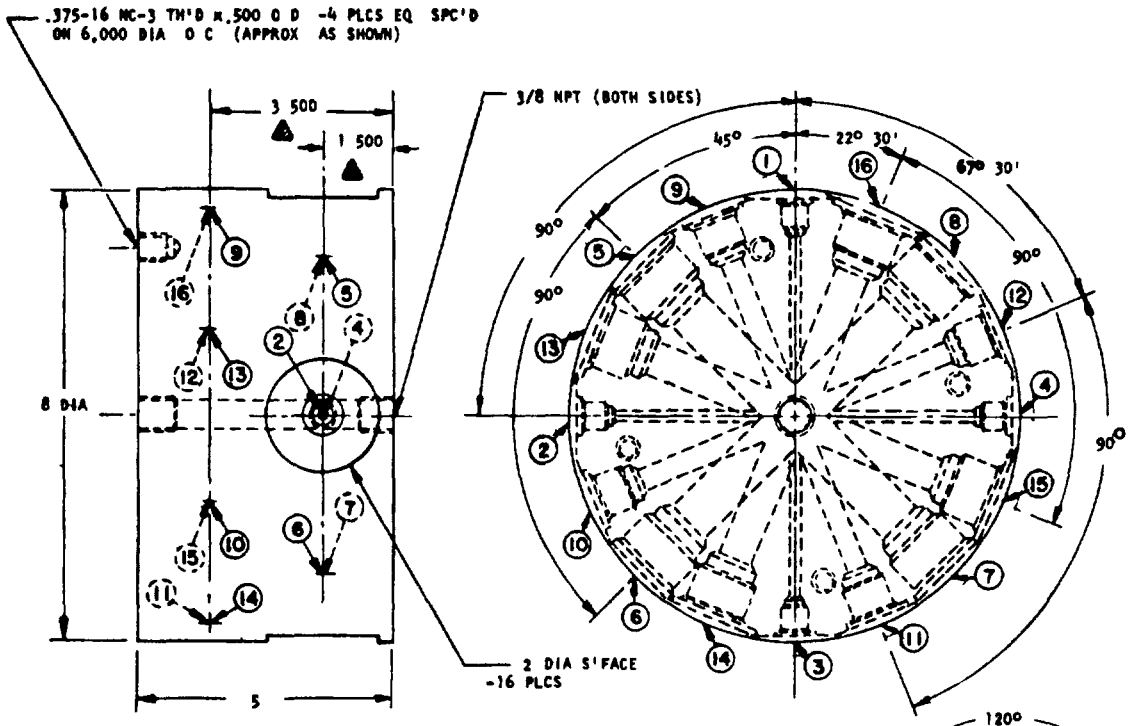
SUGGESTED METHOD OF MAKING CRUSH TEST. HYDRAULIC RIG SHOWN MAY BE MECHANICAL PROVIDED AXIAL CRUSHING LOAD CAN BE ACCURATELY DETERMINED AND MAINTAINED. HYDRAULIC PRESSURE TO COMPRESS PACKING = $P = \frac{F \times N}{A}$

WHERE P = PRESSURE, PSI:
AND N = NO. MS -16 PACKINGS IN EACH LAYER
A = PISTON HEAD AREA IN SQUARE INCHES
F = 5600 LBS. FOR MS -16 PACKINGS

FIGURE 3

Crush test setup

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NOTE:

1. FINISH: COMMERCIAL CAD PLATE LESS THAN 0002 MAX (PER SURFACE)
2. DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED

	BOSS	THREAD T	A +.002 -.000	B (MIN)	C (MIN)	D +.001 +.000	E +.001 -.000	J (MIN)	P (PILOT)
▲	①&②	7/16 - 20 NF-3	.575	.438	.172	.458	089	594	385
	③&④		.562	.438	.172	454	075	.594	385
	⑤&⑥	1 5/16 -12 N-3	1.500	.688	.844	1.340	139	.938	1 225
	⑦&⑧		1.487	.688	.844	1.336	125	.938	1 225
▲	⑨&⑩	3/4 -16 NF-3	.888	.562	.391	.773	108	718	.686
	⑪&⑫		.875	.562	.391	.769	094	.718	686
	⑬&⑭	1 1/16 -12 N-3	1.347	.688	.609	1.090	139	.906	.975
	⑮&⑯		1 234	.688	609	1 086	125	.906	.975

FIGURE 4 Impulse test fixture

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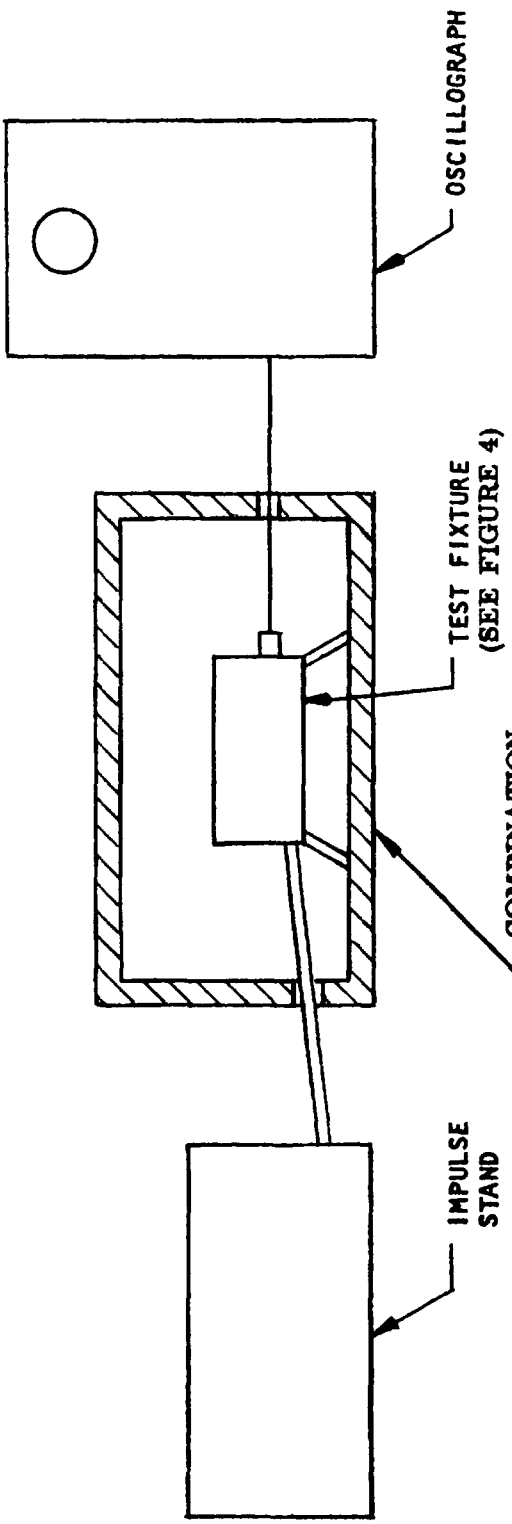
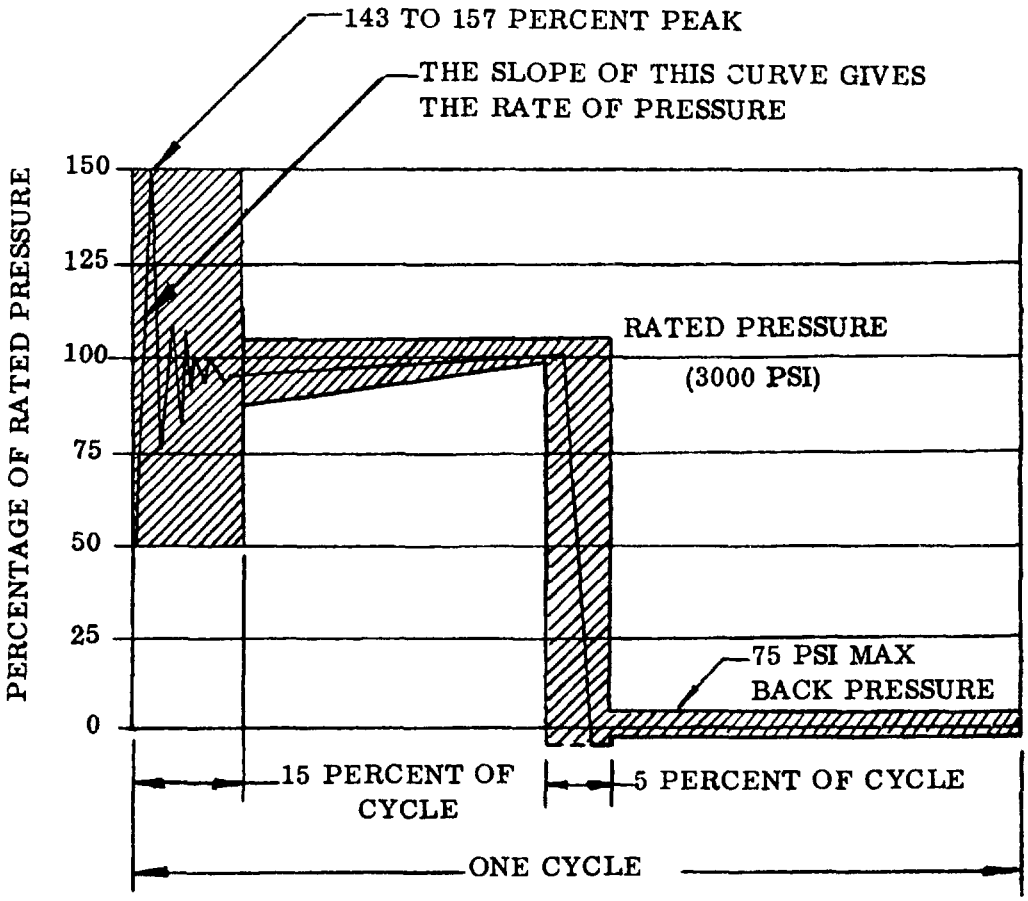


FIGURE 5

Impulse test setup

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THE CURVE SHOWN ABOVE IS THE APPROXIMATE PRESSURE TIME CYCLE DETERMINED TO BE OF PROPER SEVERITY FOR IMPULSE TESTING OF HYDRAULIC PACKINGS. ALTHOUGH IT IS MANDATORY ONLY THAT PRESSURE PEAK RISES TO 143 TO 157 PERCENT OF THE RATED PRESSURE (3000 PSI) AT SOME POINT PRIOR TO LEVELING OFF AT RATED PRESSURE, IT IS CONSIDERED HIGHLY DESIRABLE THAT THE PRESSURE TIME CURVE BE CONFINED TO THE SHADED AREA INDICATED. ONE VERY DESIRABLE BENEFIT TO BE GAINED IN THIS MANNER IS THAT RESULTS OF TESTS PERFORMED ON DIFFERENT TEST MACHINES WILL BE MORE NEARLY COMPARABLE.

FIGURE 6. Graph of pressure impulse cycle for impulse test.

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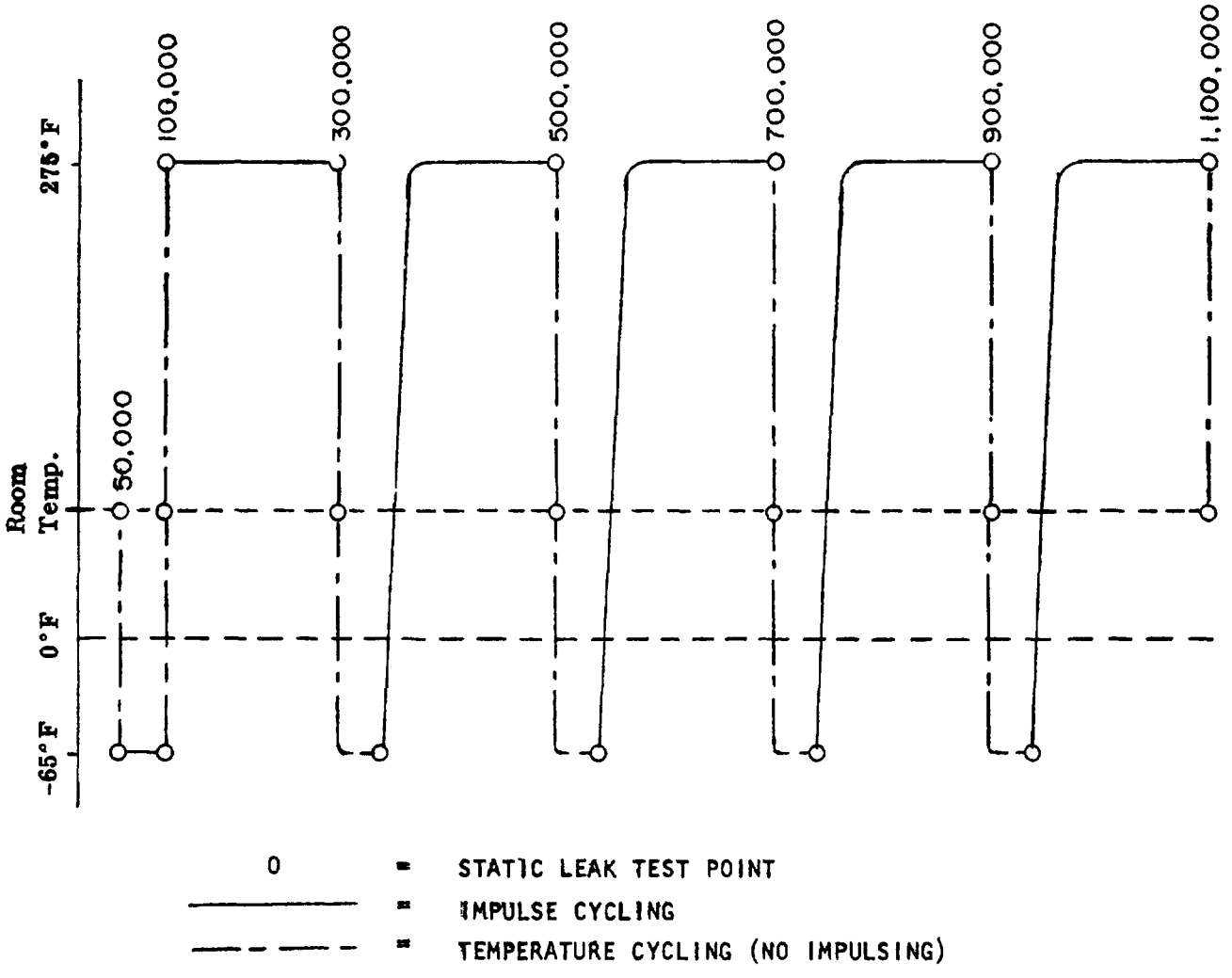


FIGURE 7

Impulse cycling schedule

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-P-81716 PACKING, PREFORMED, STRAIGHT THREAD TUBE FITTING BOSS, TYPE II

ORGANIZATION *(Of submitter)*	CITY AND STATE HYDRAULIC

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO 15419

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WESO-CODE X
NAVAL AIR ENGINEERING CENTER
PHILA., PA. 19112

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